Sum-of-Squares optimization with JuMP in Julia

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POEMA Learning Week 2

Background

Examples

Extensions

Background

Nonnegative polynomial into sum of squares (SOS)

(x^3, x^2, x) — *not* unique

$$p(x) = X^\top Q X$$

$$x^6 - 2x^4 + 2x^2 = X^\top \begin{pmatrix} 1 & 0 & -1 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 2 \end{pmatrix} X$$

$$p(x) \geq 0 \ \forall x \impliedby Q \succeq 0$$

cholesky ↓

$$x^2 + (-x^3 + x)^2 \longleftarrow X^\top \begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix}^\top \begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix} X$$

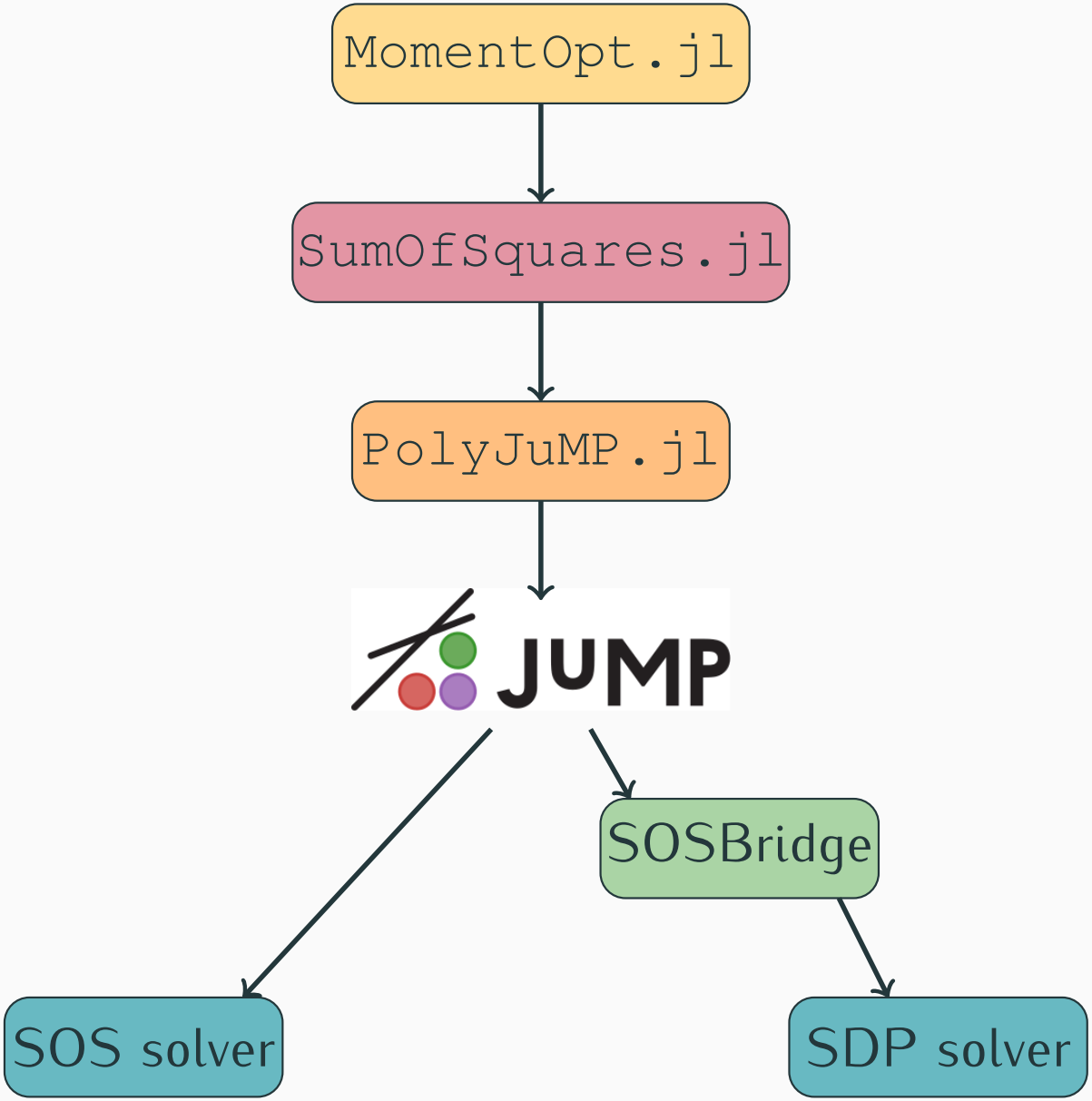
Polynomial optimization in Julia

- ahmadreza-marandi/Polyopt.jl
- rdeits/Mayday.jl
- ameliaperry/SumOfSquaresOptimization.jl
- JulieSliwak/MathProgComplex.jl
- jump-dev/PolyJuMP.jl, jump-dev/SumOfSquares.jl
- lanl-ansi/MomentOpt.jl
- tweisser/PolyPowerModels, tweisser/PolyModels
- wangjie212/TSSOS, wangjie212/NCTSSOS
- wangjie212/SONCSOCP
- blegat/CondensedMatterSOS.jl

History

- Started in August 2016: `SOSTOOLS` demos.
- October 2016: Amelia Perry, Joey Huchette, Robin Deits and Chris Coey join, package split into `MultivariatePolynomials`, `PolyJuMP` and `SumOfSquares`.
- April 2017, `DynamicPolynomials`, `TypedPolynomials`,
- August 2017 `MultivariateMoments` and `SemialgebraicSets` factored out of `SumOfSquares`.
- 2017–2019 `MathOptInterface` (MOI) → rewrite of `JuMP`, `PolyJuMP` and `SumOfSquares`.
- 2019–2020 `MutableArithmetics` for performance.
- 2019–2020 Tillmann Tweisser joins, develop `MomentOpt`.
- 2020–2021 Marek Kaluba joins, develop `SymbolicWedderburn` for symmetry reduction.

Overview


MomentOpt.jl
SumOfSquares.jl
PolyJuMP.jl
JUMP
SOSBridge
SOS solver
SDP solver

Examples

Polynomial Optimization

https://jump.dev/SumOfSquares.jl/stable/generated/Polynomial%20Optimization/polynomial_optimization/

Noncommutative variables

https://jump.dev/SumOfSquares.jl/stable/generated/Noncommutative%20and%20Hermitian/noncommutative_variables/

Symmetry reduction

- https://jump.dev/SumOfSquares.jl/stable/generated/Symmetry/permutation_symmetry/
- https://jump.dev/SumOfSquares.jl/stable/generated/Symmetry/even_reduction/
- https://jump.dev/SumOfSquares.jl/stable/generated/Symmetry/dihedral_symmetry_of_the_robinson_form/

Extensions

New polynomial implementation

https://blegat.github.io/CondensedMatterSOS.jl/dev/generated/Ising/

New certificate

`https://jump.dev/SumOfSquares.jl/stable/generated/Extension/certificate/`

If certificate not flexible enough: define new MathOptInterface bridge.

New solver

https://jump.dev/SumOfSquares.jl/stable/generated/Extension/univariate_solver/

PolyJuMP extension

What does this means ?

```
@polyvar x y
@constraint(model, x^4 + y^4 >= x * y - 1)
```

PolyJuMP needs a polymodule (shortcut: `SOSModel`):

```
model = Model()
setpolymodule!(model, SumOfSquares)
```

With `SumOfSquares.jl`, it is interpreted as:

```
@constraint(model, x^4 + y^4 - x * y + 1 in SOSCone())
```

New bases

MultivariateBases provides:

- `FixedPolynomialBasis`: Any vector of polynomials.
- Monomial bases: `MonomialBasis` and `ScaledMonomialBasis`.
- Orthogonal bases:
  - Hermite bases: `ProbabilistsHermiteBasis` and `PhysicistsHermiteBasis`.
  - `LaguerreBasis`.
  - Gegenbauer bases:
    - LegendreBasis.
    - Chebyshev bases: `ChebyshevBasisFirstKind` and `ChebyshevBasisSecondKind`

Any other basis implementing the MultivariateBases API can be used with SumOfSquares!

New algebraic sets

See `https://jump.dev/SumOfSquares.jl/stable/generated/Extension/hypercube/`.

Conclusion

- `MathOptInterface.jl` (MOI) is extensible by design: Defining new set and bridges is the way to go.
- `JuMP.jl` macros are designed for extensions as well:
- `PolyJuMP.jl` extends JuMP macros for polynomials, defines MOI set and bridges for polynomial equality in algebraic set.
- `SumOfSquares.jl` is a PolyJuMP's polymodule, defines MOI set and bridges (parametrized by certificates) for SOS constraints in basic semialgebraic set.